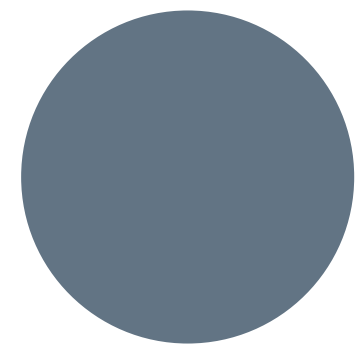
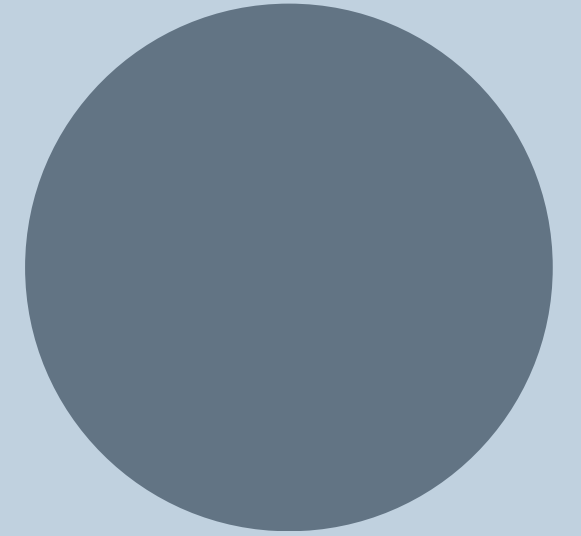

TRAFFIC AND ACCIDENT DETECTION SYSTEM



25-26J-174



MEET OUR TEAM



Pushpamal K P N Wathsalavi K HS

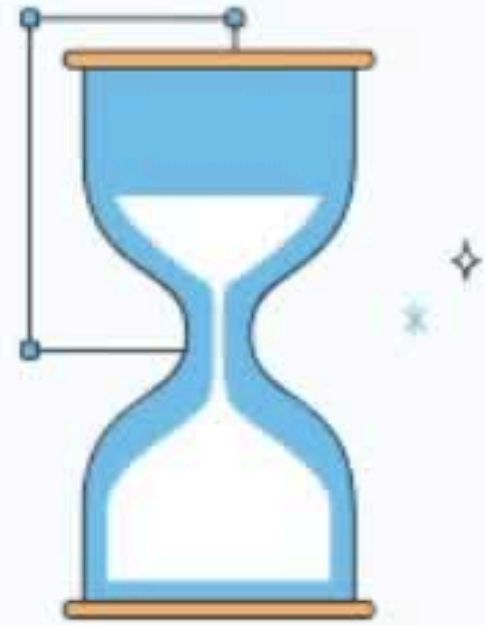


Shehara I.G.D.S



Rashmi B.G.R





TRAFFIC AND ACCIDENT DETECTION SYSTEM



25-26J-174

Date 
March 10, 2026

PROBLEM STATEMENT & STAKEHOLDERS

Context

Road traffic accidents and traffic rule violations are a major problem in many cities. Manual monitoring by traffic police is difficult and cannot cover all areas all the time.

Stakeholders

- Traffic Police
- Road Users (drivers and riders)
- Government / Road Safety Authorities
- Emergency Services (Hospitals, Ambulance)

Observed Issue

- Many traffic violations happen without being detected
- Slow accident detection delays emergency response.
- Lack of transparency in violation appeal processes.

Why It Matters

- Prevent to more road accidents and injuries.
- Delays in emergency response can increase damage and loss of life.
- Inefficient violation management wastes time and resources.

RESEARCH GAP & OBJECTIVES

● Research Gap

Existing traffic monitoring systems mainly focus on detecting only one type of violation.

Most systems do not combine violation detection, accident detection, emergency alerts, and citizen appeal management into one integrated platform.

● Main Objective

To develop an AI-based Traffic Violation and Accident Detection System that can:

- Detect multiple traffic violations automatically
- Identify vehicle number plates of violating vehicles
- Detect road accidents and assess accident severity
- Send emergency alerts to authorities quickly
- Provide a transparent system for citizens to submit appeals

This system aims to improve road safety and make traffic management more efficient.

PROPOSED SOLUTION OVERVIEW

● Solution Overview

- An AI traffic monitoring system that detects rule violations, reads vehicle numbers, manages appeals, and identifies accidents with severity assessment while alerting police and hospitals.

Key Features

- FR1- Detects red-light and seatbelt violations.
- FR2- Spots riders without helmets and number plates.
- FR3- Lets people submit appeals with AI prediction.
- FR4- Detects accidents, checks severity, and sends alerts.

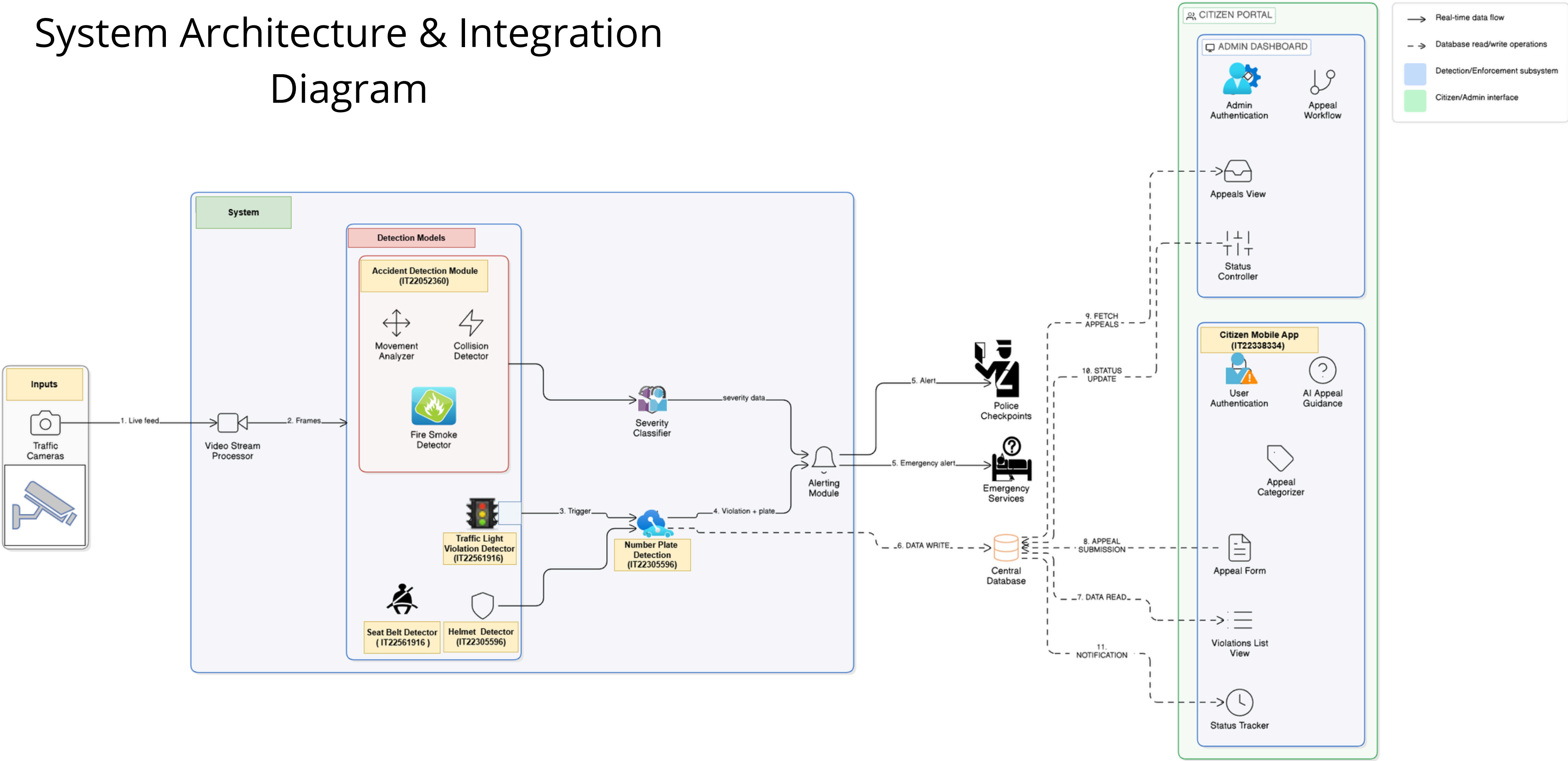
Non-Functional Focus

- Performance: Detects traffic violations quickly with AI.
- Reliability: Sends accurate automatic alerts.
- Scalability: Works with many cameras.
- Security: Keeps records and vehicle data safe.

Current Completion Status

- All four analytical modules implemented
- System architecture integrated with data pipelines and ML models
- Overall project completion: ~90%

System Architecture & Integration Diagram



COMMERCIALIZATION / IMPACT

Target Users / Market

- Traffic Police Departments
- Smart City Traffic Management Systems
- Government Road Safety Authorities

Value Proposition

- Automatic detection of traffic violations
- Faster accident detection and emergency alerts
- Improved transparency through citizen appeal system
- Helps authorities monitor traffic more efficiently

Cost & Feasibility

- Uses existing traffic cameras and AI models
- Can be deployed on cloud or local servers
- Scalable for city-wide traffic monitoring

Adoption Plan / Next Steps

- Integration with police monitoring systems
- Future expansion to multiple cities
-

[← Back to Dashboard](#)

Violation Alerts

Monitor and manage violation notifications and alerts

🔔 Recent Alerts (4)				
VIOLATION REF ID	LICENSE PLATE	OWNER SMS	AUTHORITY ALERT	TIME STAMP
TV-2024-001	ABC-1234	Delivered	Sent	2024-01-15 14:30:25
TV-2024-002	XYZ-5678	Pending	Sent	2024-01-15 15:45:12
TV-2024-003	DEF-9012	Failed	Failed	2024-01-15 16:20:33
TV-2024-004	GHI-3456	Delivered	Sent	2024-01-15 17:10:45

AI Appeal Outcome Predictor

Estimate the likelihood that a new appeal will be approved before you submit.

Appeal Reason (full text)

Describe what happened, any evidence, and why you believe the fine should be reconsidered.

The explanation below will be based only on what you write here, not on length alone.

Appeal Reason Length: 0 characters

Violation Type

Speeding

Violation Date

09/03/2026

Used to derive weekday & month

Evidence Uploaded

Yes

Previous Violations

0

Automatically calculated from your past appeals.

Days After Violation

3

Automatically calculated from violation date.

Predict Approval Probability

AI Appeal Reason Categorizer

Automatically categorize your appeal reason (e.g. Incorrect Violation, Emergency Situation, Technical Error) before you submit.

Appeal Reason (full text)

Paste or type your appeal reason here to see how the system categorizes it.



**Thank
You**